Conditional expectation

▽ Let A be some event, then

$$E[g(X) \mid A] = \begin{cases} \sum_{\forall x} g(x) \cdot P_{X|A}(x) \\ \int_{-\infty}^{\infty} g(x) \cdot f_{X|A}(x)\, dx \end{cases}$$

▽ Ex. Let (X, Y) have joint PDF:

$$f_{X,Y}(x,y) = \begin{cases} 60\, e^{-4x-6y}, & 0<x<y<\infty \\ 0, & \text{o.w.} \end{cases}$$

Find $E(Y \mid X = x)$.

- Need to compute $\int y \cdot \underbrace{f_{Y|X=x}(y)}_{?}\, dy$

$$\bullet\ f_{Y|X=x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

$$= \frac{60\, e^{-4x-6y}}{\int_x^{\infty} 60\, e^{-4x-6y}\, dy}$$

$$= \frac{\cancel{60}\, e^{\cancel{-4x}-6y}}{\cancel{60}\, \cancel{e^{-4x}} \cdot \frac{1}{6} e^{-6x}}$$

$$= 6\, e^{-6y+6x}, \quad 0<x<y<\infty$$

$$\therefore\ E(Y|X=x) = \int_x^{\infty} y \cdot 6\, e^{-6y+6x}\, dy$$

$$= 6 \cdot e^{6x} \cdot \int_x^{\infty} y\, e^{-6y}\, dy$$

Recall: $\int u \cdot v' = u \cdot v - \int u' \cdot v$

Let $u = y \qquad v' = e^{-6y}$

$u' = 1 \qquad v = \frac{1}{-6} e^{-6y}$

$$\int_x^\infty y e^{-6y} dy = y \cdot \frac{1}{-6} e^{-6y} \Big|_x^\infty$$

$$+ \int_x^\infty \frac{1}{6} e^{-6y} dy$$

$$= 0 - \left(\frac{1}{-6}\right) x \cdot e^{-6x} + \frac{1}{6} \cdot \frac{1}{6} e^{-6x}$$

$\therefore E(Y|X=x) = x + \frac{1}{6}, \quad x > 0.$

- NOTE : $E(Y \mid X = x) = x + \frac{1}{6} = h(x)$

- $E(Y \mid X) = h(X)$ which is a RV !

- $E(h(X)) = \int_{-\infty}^{\infty} h(x) \cdot f_X(x)\, dx$

$$= \int_{-\infty}^{\infty} \boxed{\int_{-\infty}^{\infty} y \, f_{Y \mid X = x}(y)\, dy} \, f_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y \, f_{X,Y}(x, y)\, dy\, dx$$

$$= \int_{-\infty}^{\infty} y \overset{f_Y(y)}{\left[\int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx \right]} dy$$

$$= E(Y)$$

$$\therefore \; E[E(Y \mid X)] = E(Y)$$

Def. The conditional variance of X given A is

$$Var(X \mid A) = E((X - E(X \mid A))^2 \mid A)$$

$$= E(X^2 \mid A) - [E(X \mid A)]^2$$

$$E[Var(Y \mid X)] + Var(E(Y \mid X)) = Var(Y)$$

The proof is omitted, the formula is very useful.